UdbhavaX'26

- *Project Title:Plastic Waste Recycle Tracking System*

  Domain: ST-05 (Sustainability)
- *Team Name: [PATHFINDERS]*
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  Department: [Your Department, e.g., CSE]
- *Event: Udbhava X'26, Coding Club SR University*
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Problem Scenario

The Real-World Situation:

Cities like Delhi, Mumbai, and Chennai generate huge amounts of plastic waste daily.
Households mix plastic with general waste
Waste collectors lack digital tracking tools
Recycling centers receive inconsistent supply
Government lacks accurate recycling data

The Current Failure:

1. **No Waste Segregation at Source**
   Households do not separate plastic waste properly.
2. **Lack of Tracking System**
   No proper mechanism to track plastic collection and recycling.

Current Pain Points (Why it Breaks)

- **Mixed Waste Disposal at Source:**Households often dump plastic along with organic and other waste, making segregation difficult and reducing recyclability.
- **Lack of Real-Time Tracking:**There is no system to track how much plastic waste is collected, transported, or recycled, leading to data gaps..
- **Unorganized Informal Workforce:**Waste collectors operate without digital records, proper coordination, or recognition, causing inefficiencies.
- **No Centralized Data System**
  Recycling data is scattered across different agencies, making monitoring and decision-making difficult.

Objectives & Success Criteria

Primary Objective:

To build a **smart, centralized plastic waste management system** that:

Tracks plastic waste from source to recycling

Improves segregation and participation

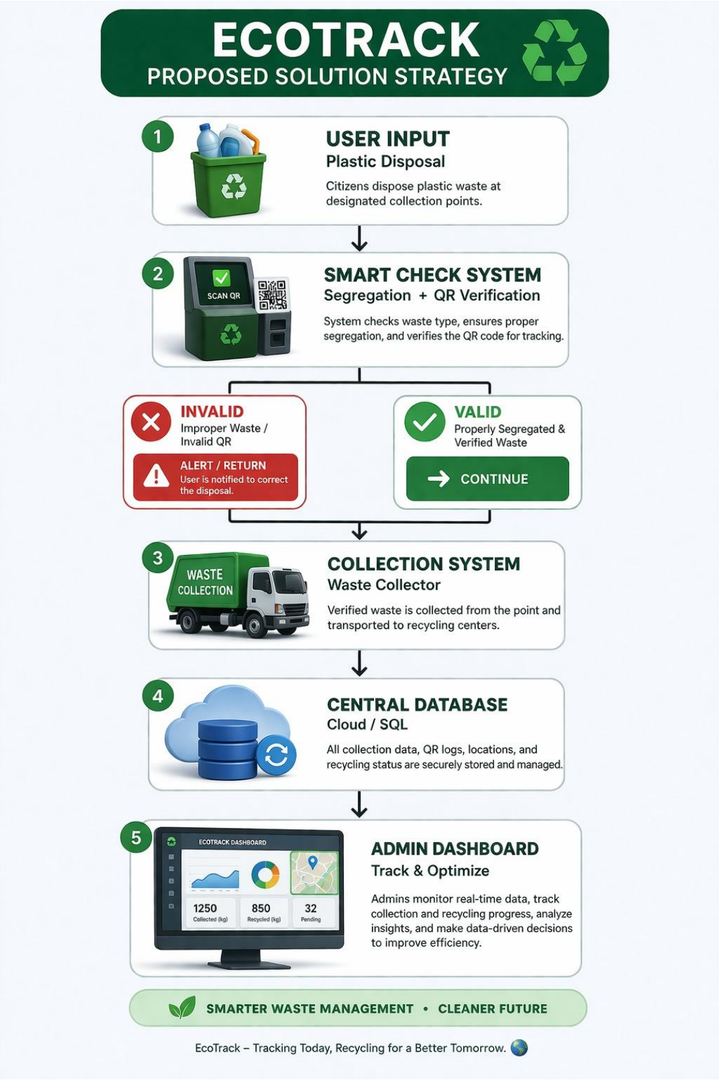
. **Success Criteria**:

- Increase in **plastic segregation rate (>70%)**Implement "Smart Logic"
- Participation of households and collectors
- Reduction in landfill plastic waste
- Availability of real-time analytics dashboard

Proposed Solution Strategy

Key Solution Strategies:

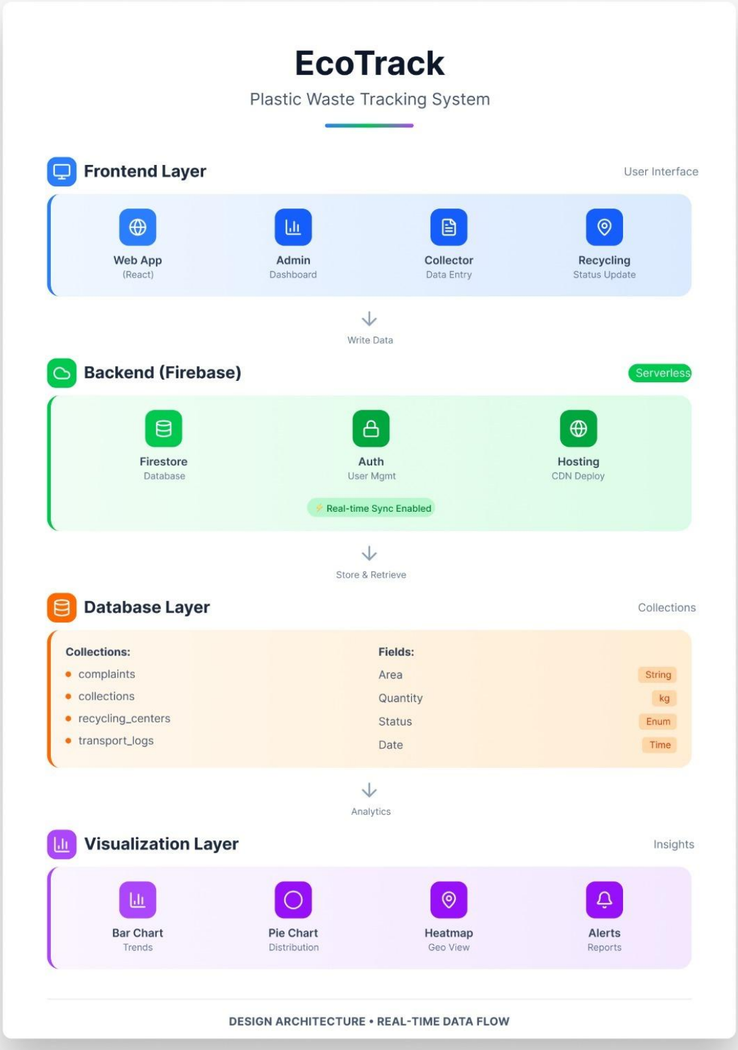
1. Mobile App for Households & Collectors
2. QR Code / RFID-based Waste Tracking
3. Cloud-based Centralized Database
4. Real-time Dashboard for Government
5. Incentive System (Rewards for Recycling)
6. Integration of Informal Workers



System Architecture

Architectural Overview (3-Tier):

- User Interface (Mobile/Web App).
- Backend Logic Layer: Server API and firestore
- Database layer: Firestore cloud storage
- Analytics Dashboard



Data Flow / Workflow

User scans QR code while disposing plastic

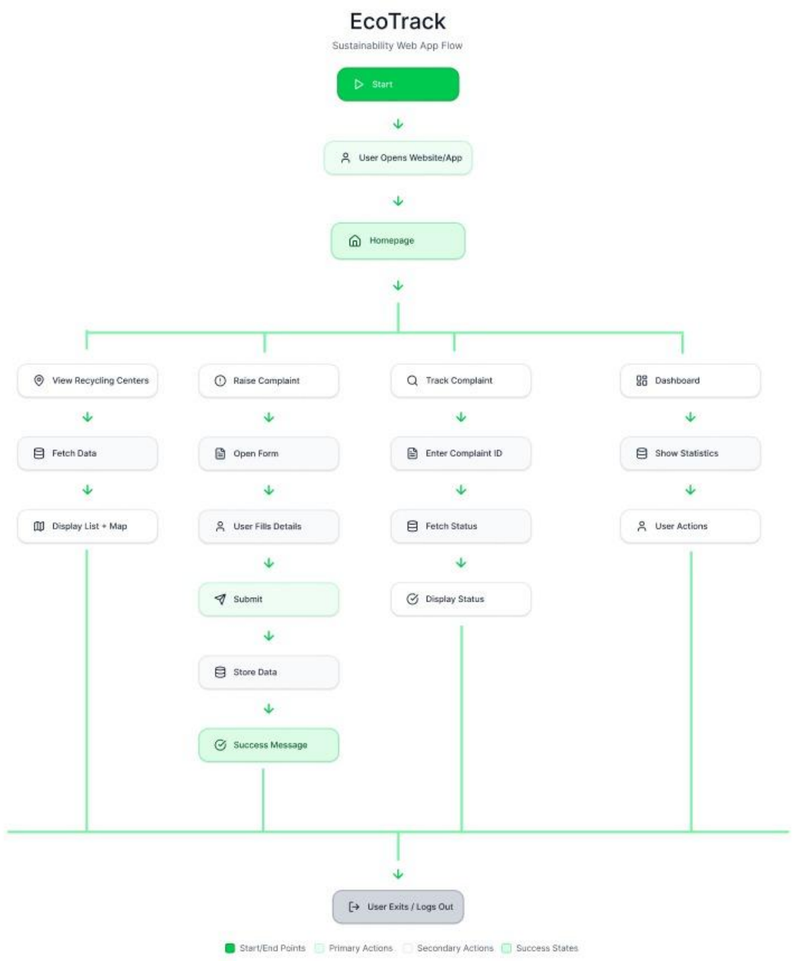
Waste collector collects and scans again

Data sent to cloud server

Recycling center updates processed data

Dashboard shows:

Collected waste
Recycled waste
Area-wise performance



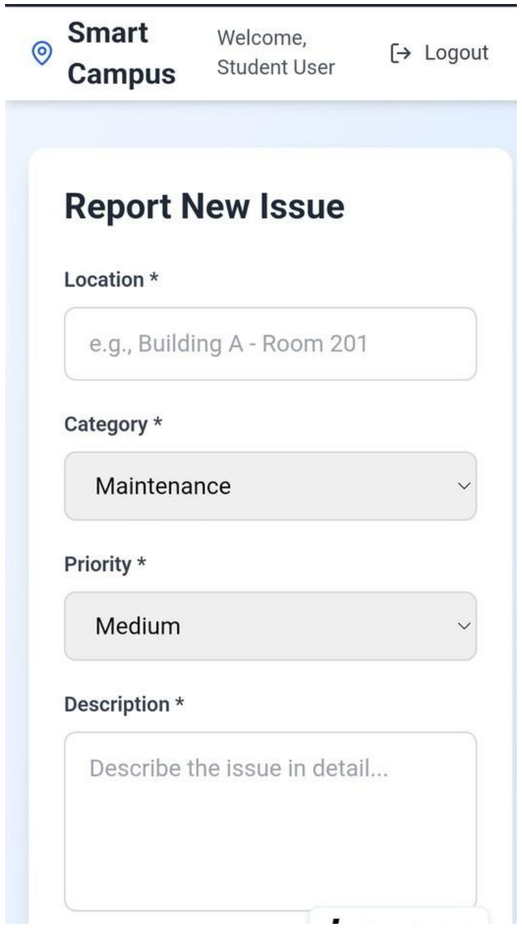
Implementation & Demo Reference

Implemented Features (Reference Prototype):

- Role-based
- Structured Reporting Form with Dropdowns.
- Admin Dashboard with Statistics Cards.
- Visual Priority Highlighting (Red badges for High Priority).

LINK REQUIREMENT:

GitHub Repo:[vyshnavibharathapu6-hub/PATHFINDERS-ECOTRACK:]



Impact & Benefits

Improved recycling efficiency
Reduction in plastic pollution
Data-driven decision making
Formal recognition for waste workers
Cleaner cities

Limitations

- Manual Data Dependency
  - System relies on manual data entry by users
  - May lead to human errors or incorrect data
- Lack of Automation
  - No integration with IoT sensors or smart bins
  - Waste tracking is not fully automatic
- Limited Real-Time Tracking
  - Does not support GPS-based transportation tracking
  - Status updates are manual
- Basic Analytics
  - Provides only simple charts and reports
  - Lacks advanced data analysis features
- Scalability Constraints
  - Designed as a prototype
  - May not handle large-scale city data efficiently

Future Scope & Roadmap

- IoT Integration
  - Use smart sensors to automatically track waste levels
  - Reduce manual data entry
- GPS Tracking
  - Enable real-time tracking of waste transportation
  - Improve monitoring accuracy
- AI & Predictive Analytics
  - Predict waste generation trends
  - Optimize recycling processes
- Mobile Application
  - Develop mobile app for collectors and authorities
  - Improve accessibility and usability